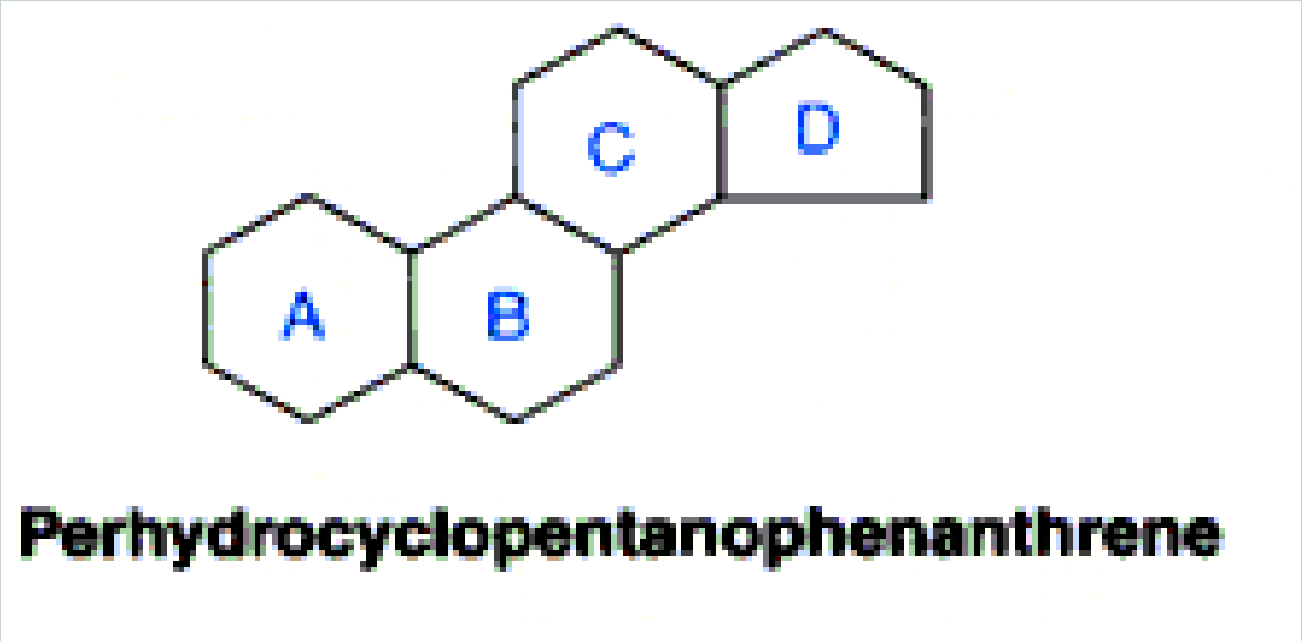
Cholesterol

DR. MUHAMMAD AHSAN

Lecture objectives

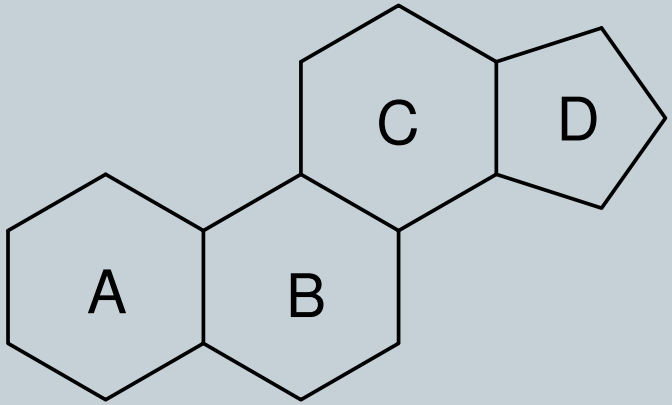
- Describe the structure of cholesterol and List its sources, precursors, functions and derivatives.
- List the steps of cholesterol biosynthesis & give the cellular and sub-cellular sites where this synthesis occurs.
- Describe the mechanism of cholesterol transport in blood

- Sterols (steroids) are class of lipids that are derivatives of a tetracyclic hydrocarbon



Steroids

- **Steroids** are lipids containing a steroid nucleus (core structure)
- The **steroid nucleus** is a fused ring system consisting of three cyclohexane rings and one cyclopentane ring
- The rings are designated A, B, C and D
- Attachment of different groups to the core steroid structure leads to a wide variety of steroid compounds, including cholesterol, bile salts and steroid hormones



CHOLESTEROL

“Cholesterol is the most highly decorated small molecule in biology. Thirteen Nobel Prizes have been awarded to scientists who devoted major parts of their careers to cholesterol.

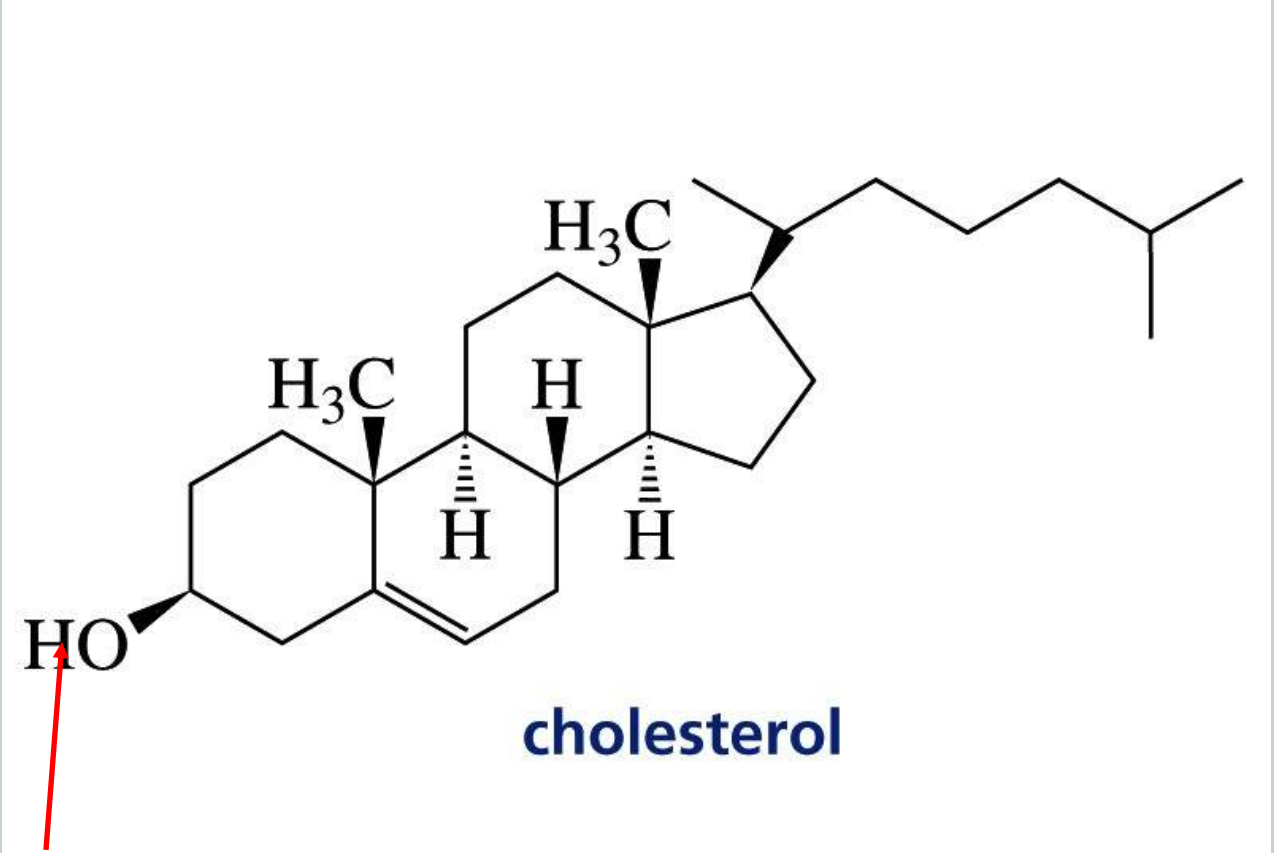
Ever since it was isolated from gallstones in 1784, cholesterol has exerted an almost hypnotic fascination for scientists from the most diverse areas of science and medicine.

The very property that makes it useful in cell membranes, namely its absolute insolubility in water, also makes it lethal.”

Overview

- Cholesterol is the most important sterols in the human body.
- It is clearly essential to life, yet its deposition in arteries is associated with cardiovascular disease and stroke, Hero or Villian?.
- In a healthy organism, an intricate balance is maintained between the biosynthesis, utilization, and transport of cholesterol, keeping its harmful deposition to a minimum

Cholesterol Structure



FA for esterification

It composed of 27 C-atoms

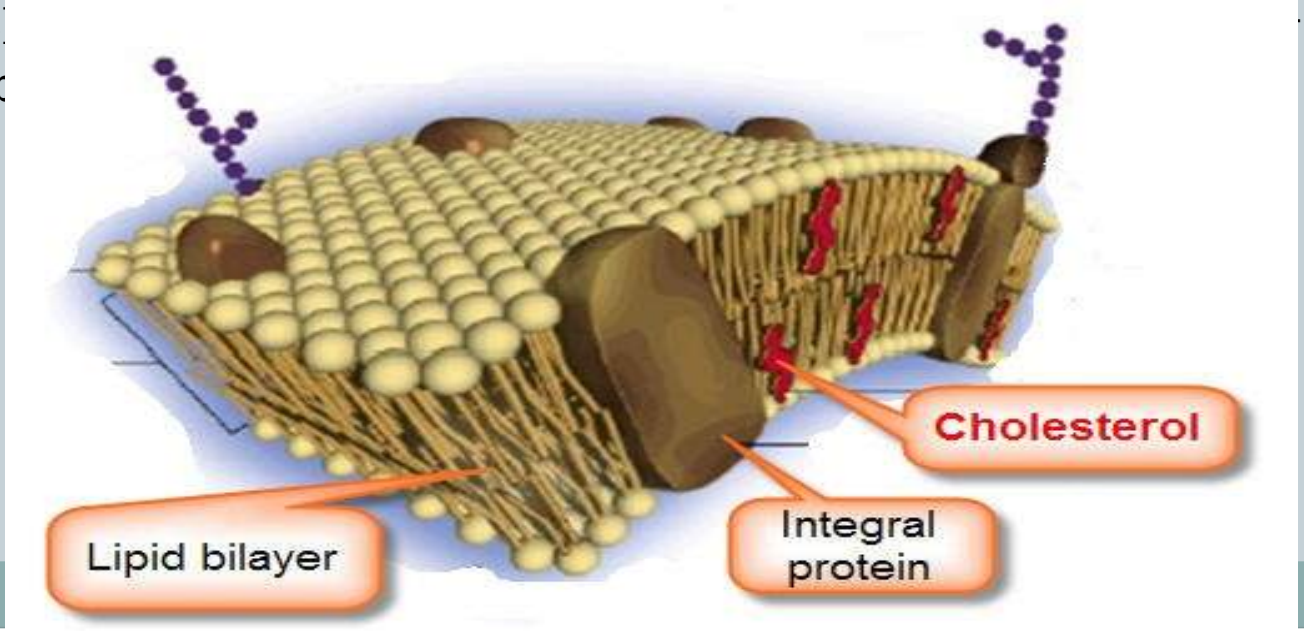
Biological importance of cholesterol

1. **Cholesterol is a crucial component of cell membranes**

- Cholesterol is compact, **rigid**, **hydrophobic** molecule with a polar OH group.

- OH group is oriented **towards the aqueous phase** and gives hydrophilic character to this end of cholesterol molecule.

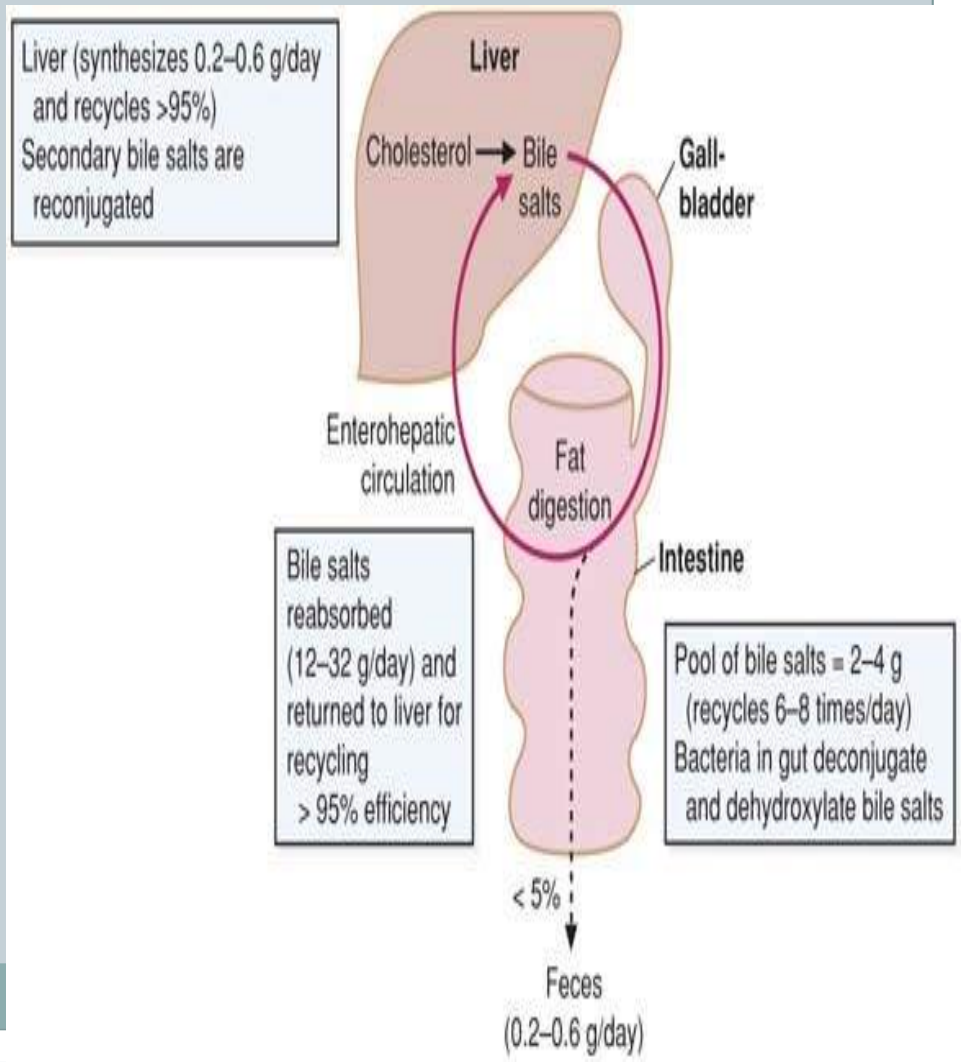
- Cholesterol : po
and **protein mob**



Biological importance of cholesterol

2. Cholesterol is the precursor of bile salts

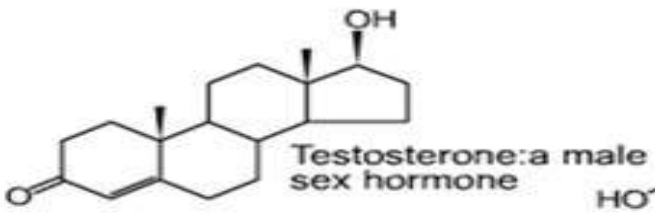
- The solubilization of hydrophobic molecules of cholesterol is aided by **bile phospholipids** and **bile salts**. They prevent cholesterol from precipitating in gallbladder in form of gallstones.
- Bile salts are **metabolites of cholesterol**.
- Cholesterol **protects gallbladder membranes** from irritating and harmful effects of bile acids and bile salts.



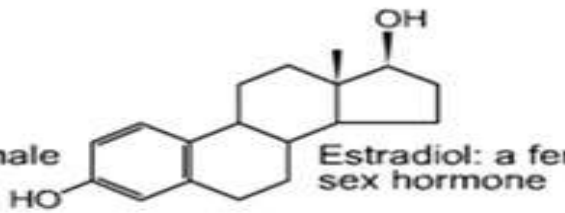
Biological importance of cholesterol

3. Cholesterol is the precursor of steroid hormones

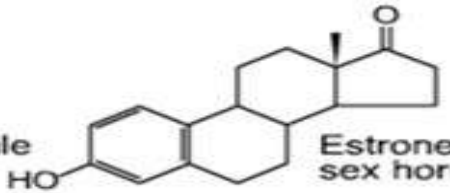
Cholesterol is the precursor of all **steroid hormones** including **estrogen, progesterone**,, **testosterone**, **corticosteroids** and



Testosterone:a male sex hormone

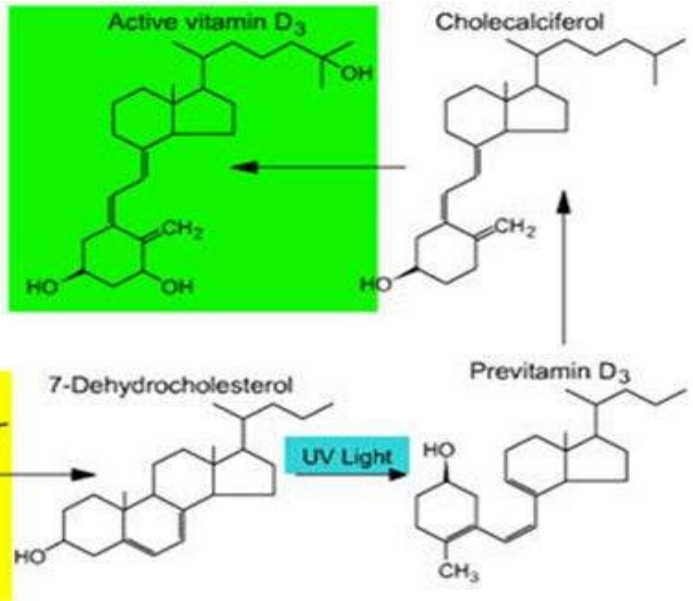


Estradiol: a female sex hormone

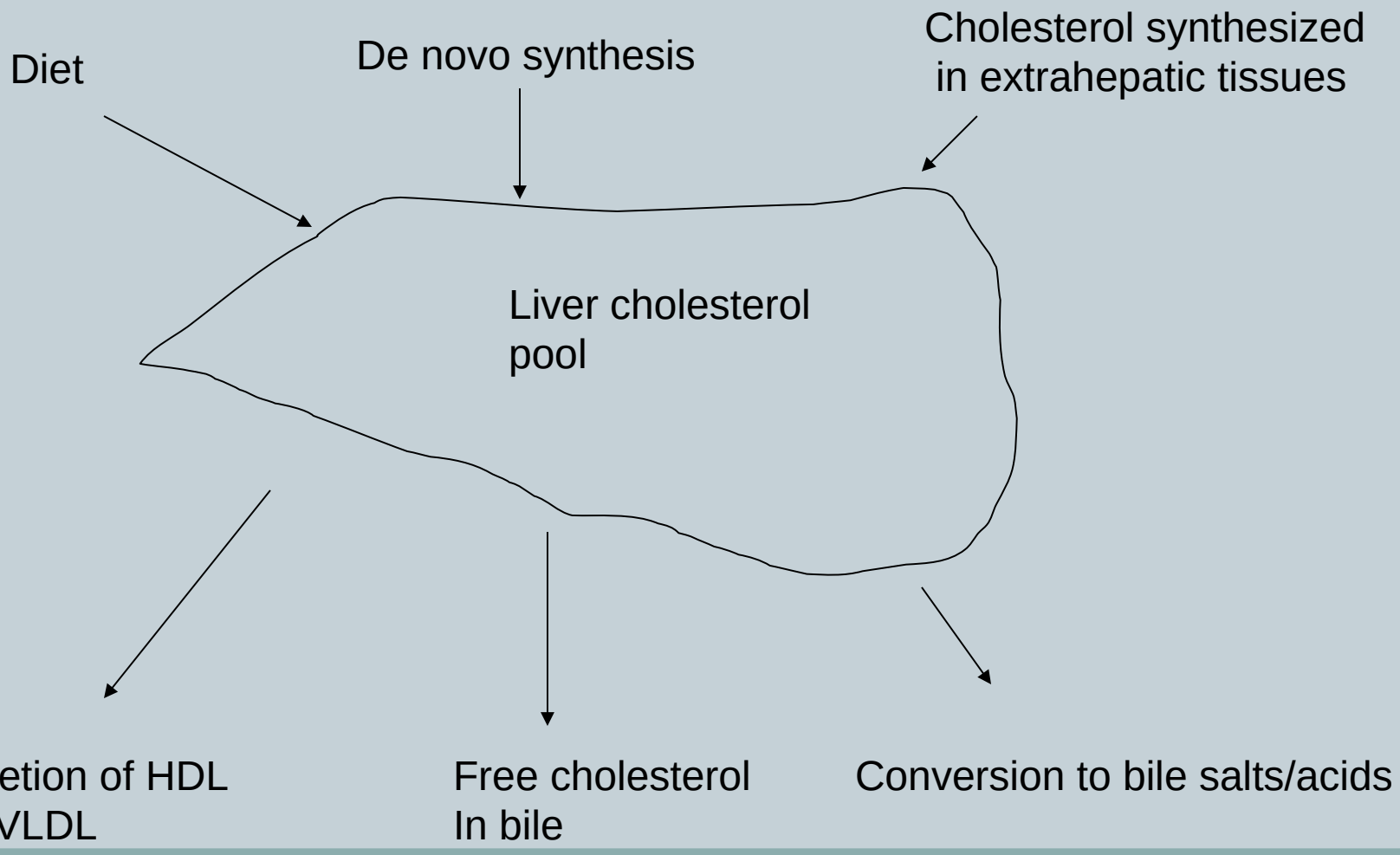


Estrone:a female sex hormone

4. Cholesterol is a precurso of vitamin D



Sources of cholesterol

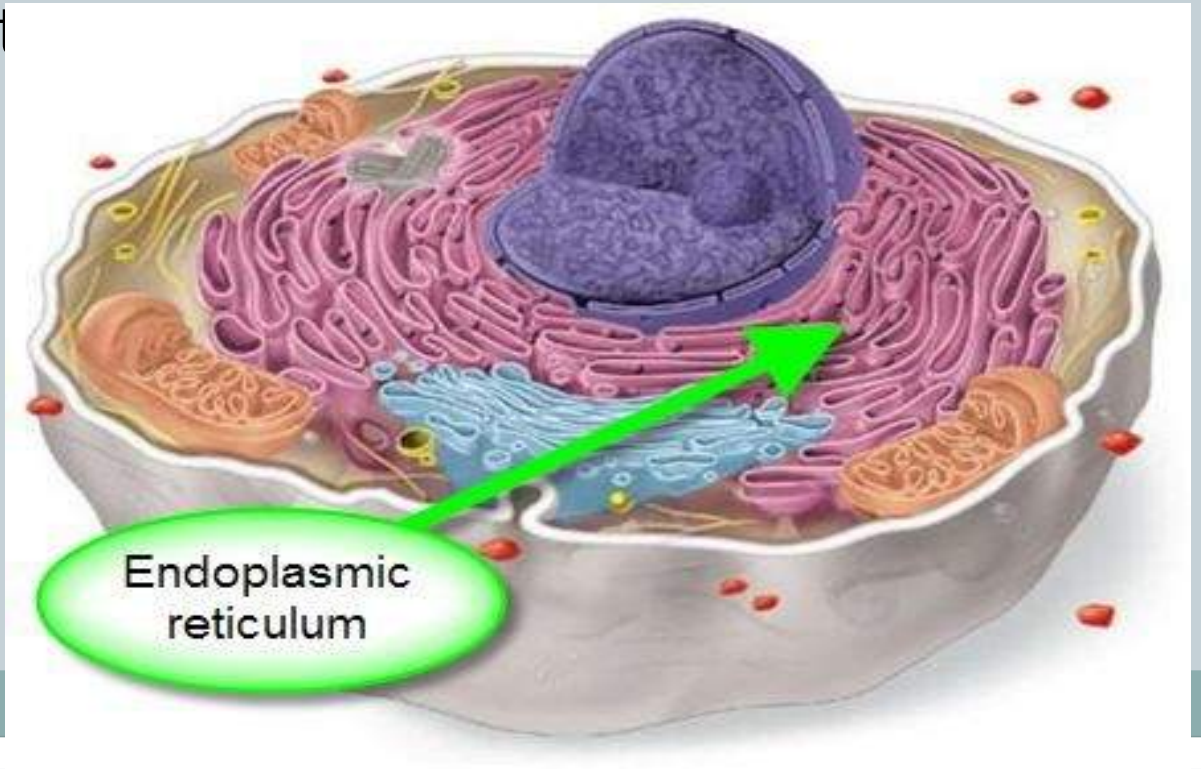


Dietary Cholesterol

- Found mainly in animal products especially egg yolks, meat, poultry, shellfish and milk.
- About 50% of dietary cholesterol is absorbed
- Increase intake = decreased absorption
- About 1 g/day is excreted.

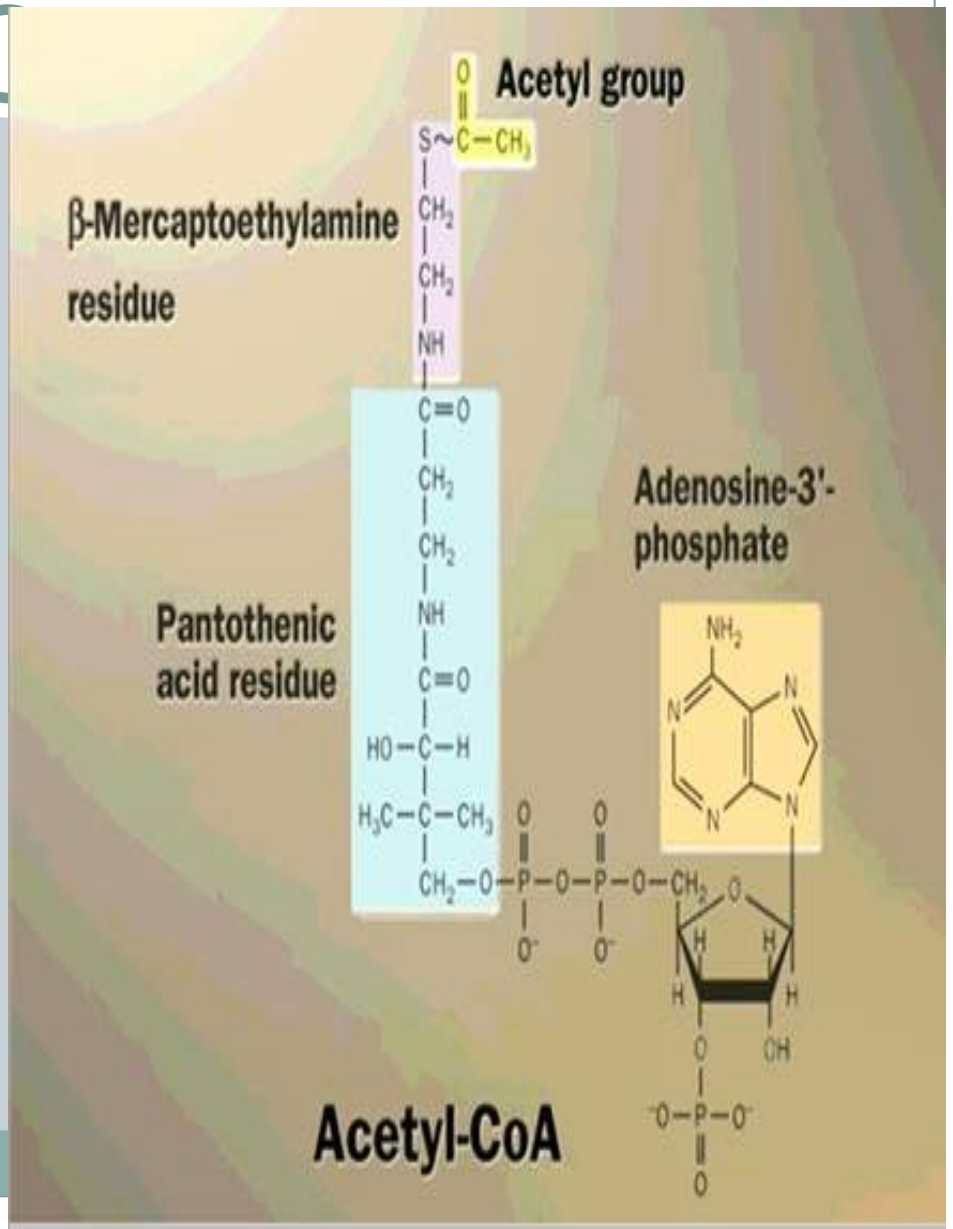
Site of cholesterol synthesis

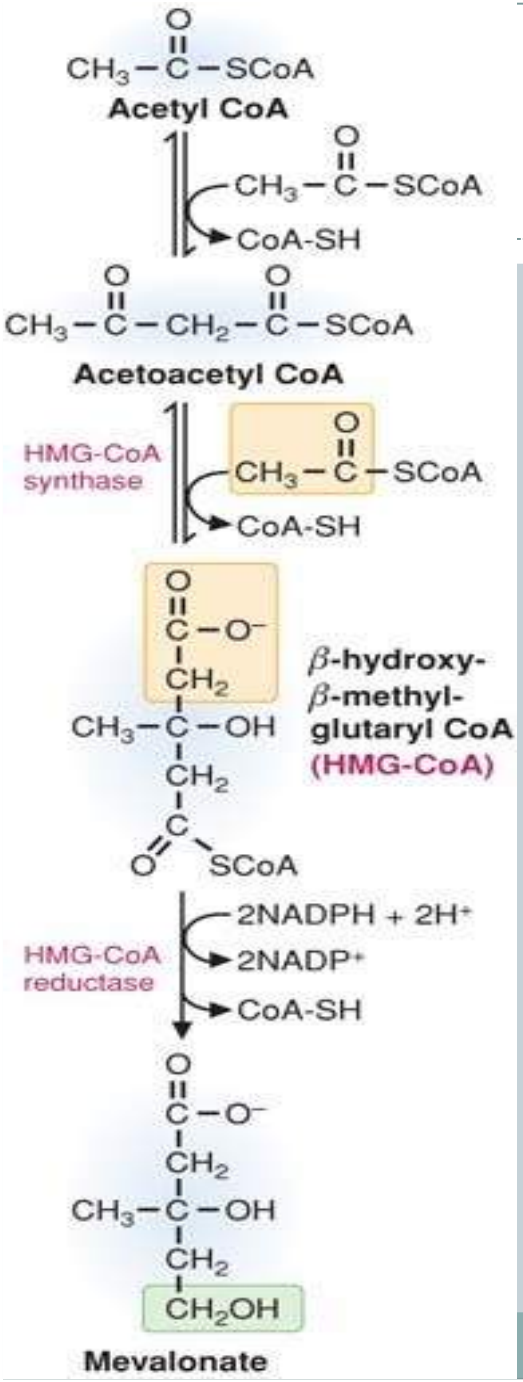
- Cholesterol is synthesized in the **cytosol** and **endoplasmic reticulum (ER).**
- **Liver** is the main site of cholesterol synthesis; also, it is synthesized in intestine, skin and other nucleated cells in t



The Substrate of cholesterol

- **Cholesterol is synthesized from acetyl groups of cytosolic acetyl coenzyme A (acetyl-CoA).**
- **Glucose and fatty acids are the major sources of acetyl-CoA**





Cholesterol Synthesis I

1) First, 2 acetyl-CoAs are condensed by **thiolase enzyme** to give acetoacetyl-CoA which condenses with another acetyl-CoA forming **hydroxymethylglutaryl CoA (HMG-CoA)**. The reaction is catalyzed by **HMG-CoA synthase**.

2) HMG-CoA is reduced to mevalonic acid (C6), this reaction requires 2NADPH. It is catalyzed by **HMG-CoA reductase** which is regulatory enzyme. This step is the **rate limiting step** in sterol biosynthesis.

Cholesterol Synthesis Stage 2: Mevalonate to 2 Activated Isoprenes

- Transfer 3 ATP to Mevalonate in order to activate C5 & OH-group of C3
- Phosphate group at C3 & Carboxyl group of C1 leave, which produces a double bound
- This allows for two active isoprenes

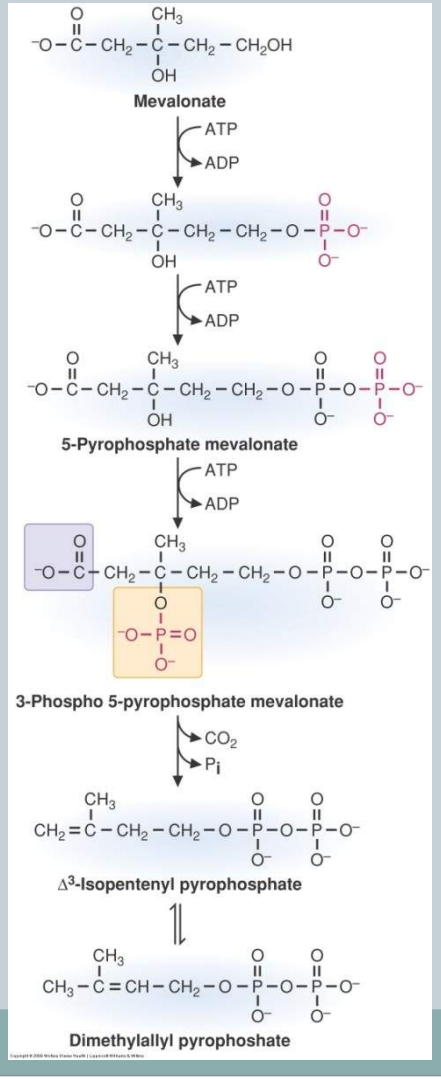


Fig.5

Cholesterol Synthesis Stage 3: Condensation of Isoprenes to for Squalene

- 1) Head to tail attachment of isoprenes to form Geranyl pyrophosphate
- 2) Head to tail condensation of Geranyl pyrophosphate and isopentenylpyrophosphate to form Farnesyl pyrophosphate
- 3) Head to head fusion of two Farnesyl pyrophosphate to form squalene

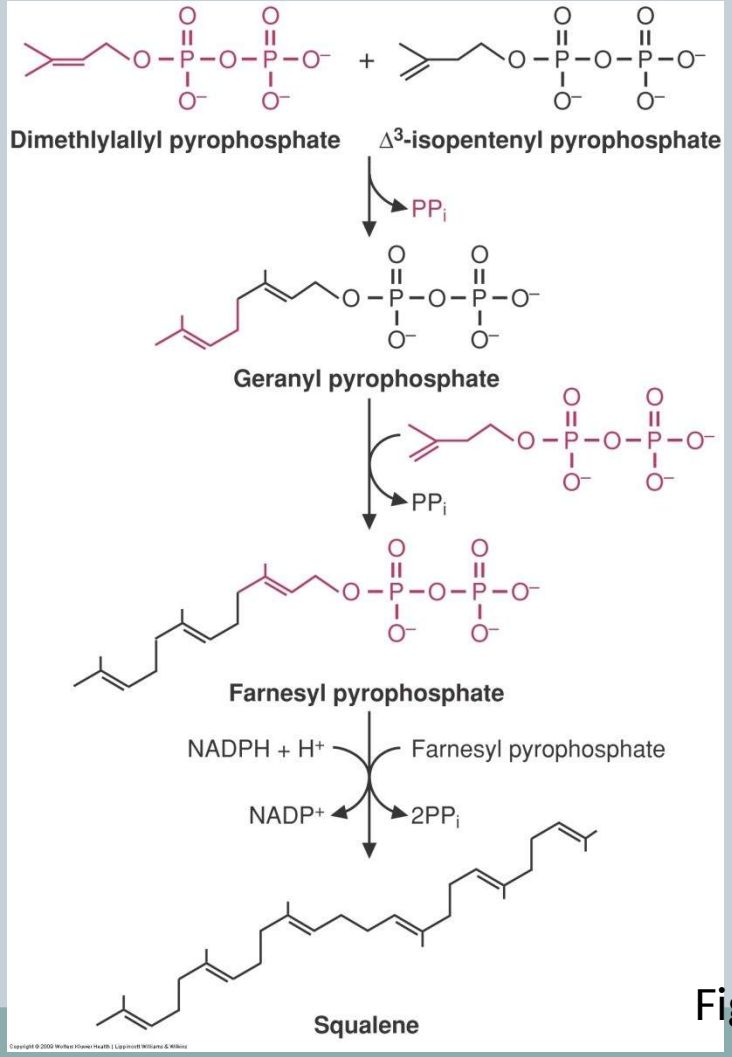


Fig.6

Cholesterol Synthesis Stage 4: Squalene to Four-Ring Steroid Nucleus

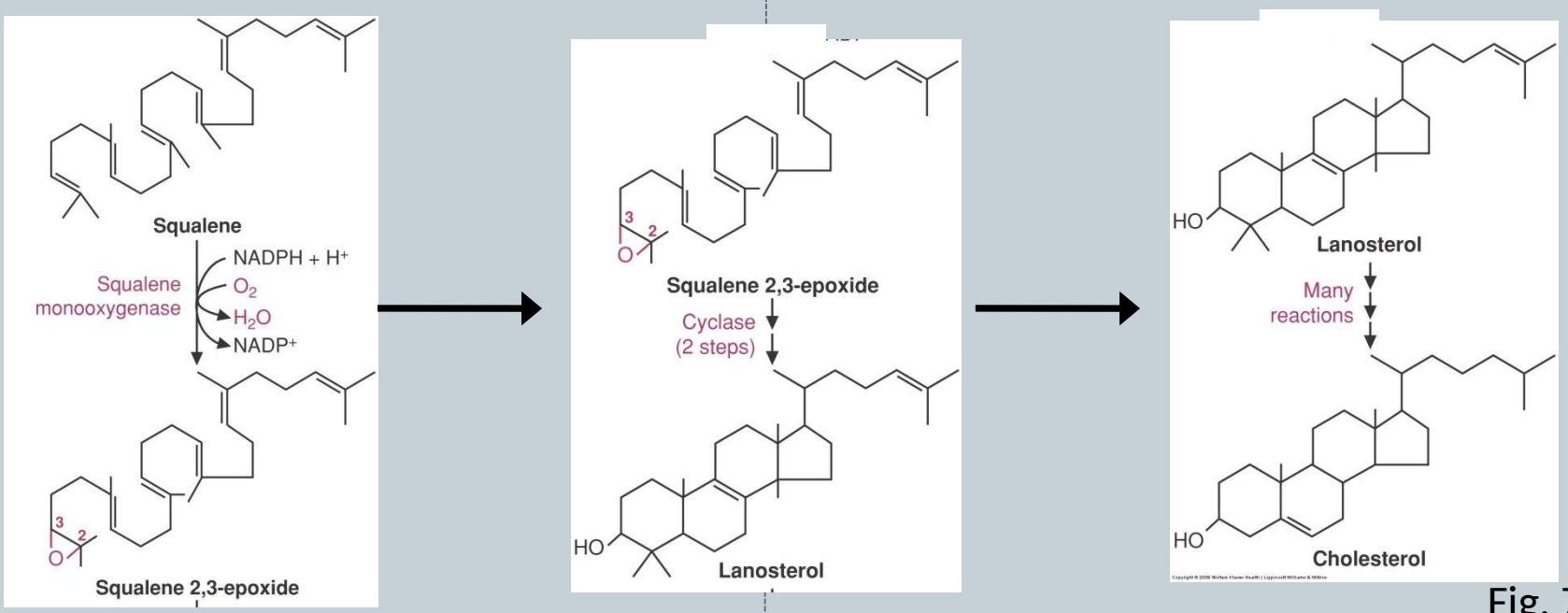
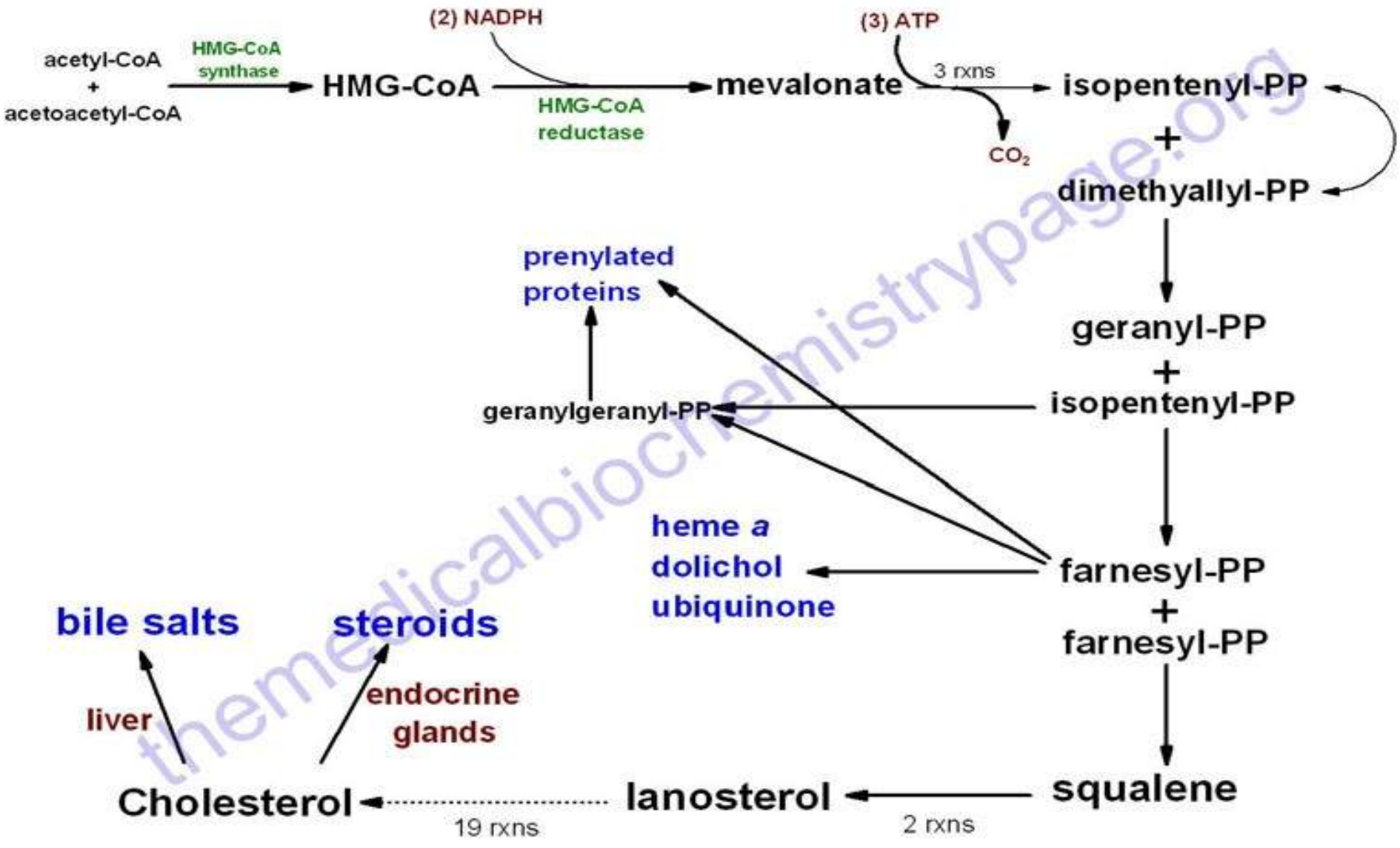


Fig. 7

- Squalene monooxygenase adds oxygen to form an epoxide
- Unsaturated carbons (double bonds) are aligned to allow cyclization and formation of lanosterol
- After many reaction get cholesterol

Cholesterol Biosynthetic Pathway



themedicalbiochemistrypage.org

Transport of cholesterol

- In plasma, 30% of cholesterol is free and 70% is in ester form. Due to the insolubility of cholesterol, their redistribution in the body requires specialized carriers capable of solubilizing and unloading them at specific target sites.
- Free cholesterol and most lipids are transported in the blood as part of soluble complexes called **lipoproteins**.
- Five main classes of lipoproteins based on their size and density called, in order of increasing density, Chylomicrons, Very-low-density lipoprotein (VLDL), Intermediate-density lipoprotein (IDL), Low-density lipoprotein (LDL) and High-density lipoprotein (HDL).

Transport of cholesterol

1) **Chylomicron** which is basically fat droplet containing little protein, pick up dietary cholesterol from the intestine. Fats are delivered to adipose tissues leaving a chylomicron remnant containing mostly cholesterol that are brought into the liver by binding with receptor that recognize ApoE in chylomicron remnant.

2) **VLDL**, formed in liver and contain excess triacylglycerol and cholesterol, transport cholesterol from liver to plasma. During transport in the bloodstream, VLDL delivers triacylglycerol to tissues leaving IDL molecules, which contain an even higher percentage of cholesterol.

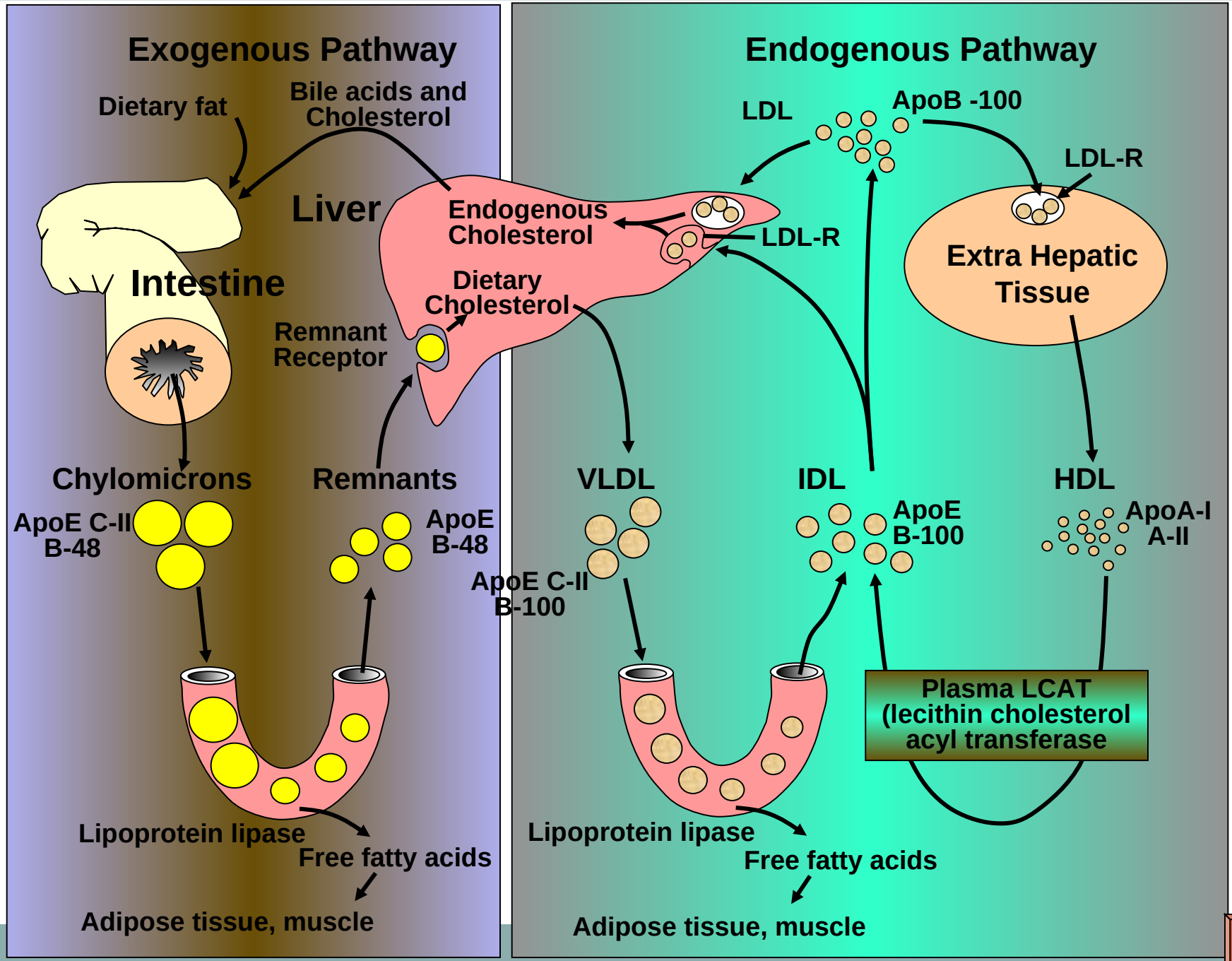
3) The IDL molecules lose triacylglycerols in the bloodstream until they form LDL molecules, which have the highest percentage of cholesterol within them, thus LDL is called “Bad cholesterol”.

Transport of cholesterol

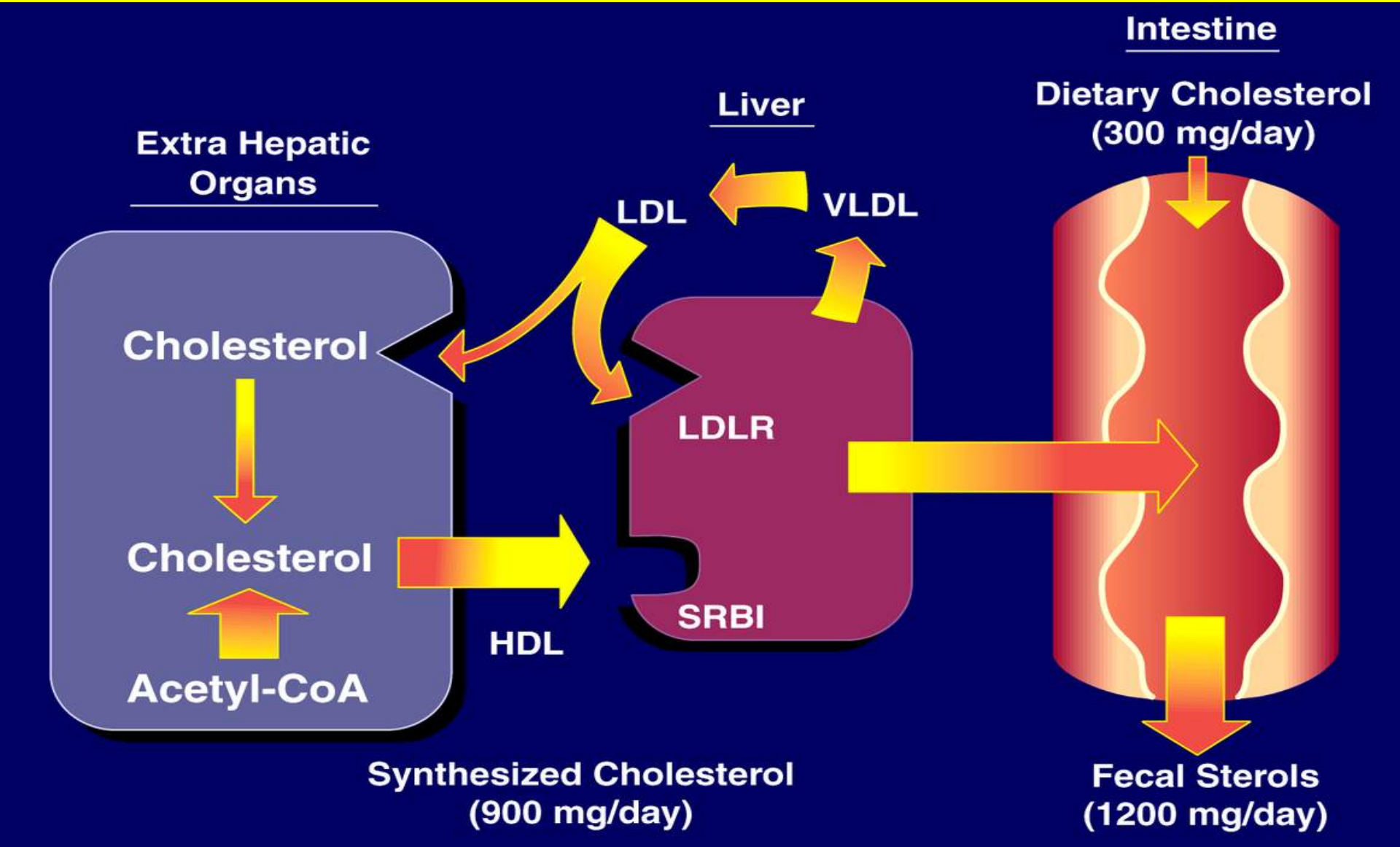
4) **LDL** is taken up by peripheral tissues through LDL receptors which recognize Apo-B100. Cholesterol is liberated and stored as cholesterol ester that produced by the action of enzyme cholesterol acyl transferase (ACAT).

5) **HDL** removes cholesterol from extrahepatic tissues and esterifying it using plasma enzyme lecithin cholesterol acyl transferase (LCAT), thus HDL is called “Good cholesterol”.

6) HDL is taken up by liver after binding with HDL receptors which recognize Apo-A1, and hydrolyzed to liberate cholesterol. Released cholesterol either enters in the structure of other lipoproteins, bile salts or excreted in bile.


Exogenous Pathway
Dietary fat
Bile acids and Cholesterol
Liver
Intestine
Endogenous Cholesterol
Dietary Cholesterol
Remnant Receptor
Chylomicrons
ApoE C-II B-48
Remnants
ApoE B-48
Lipoprotein lipase
Free fatty acids
Adipose tissue, muscle
Endogenous Pathway
LDL
ApoB -100
LDL-R
LDL-R
Extra Hepatic Tissue
VLDL
ApoE C-II B-100
IDL
ApoE B-100
HDL
ApoA-I A-II
Plasma LCAT (lecithin cholesterol acyl transferase
Lipoprotein lipase
Free fatty acids
Adipose tissue, muscle

Cholesterol balance



Bile acids & bile salts

- The end products of cholesterol utilization are the bile acids, synthesized in the liver by oxidation.
- Synthesis of bile acids is one of the predominant mechanisms for the excretion of excess cholesterol.
- The most abundant bile acids in human bile are **chenodeoxycholic acid** and **cholic acid** that are identified as primary bile acids.

Hypercholesterolemia

- Elevated level of blood cholesterol; higher concentrations of LDL and lower concentrations of functional HDL are strongly associated with cardiovascular disease because these promote athersclerosis which leads to myocardial infarction and stroke.
- The normal range for total blood cholesterol is 140 to 200 mg/dl. The total number doesn't tell the whole story, because there are two types of cholesterol -HDL and -LDL.
- **Familial hypercholesterolemia** is a genetic disorder characterized by high cholesterol level. Patients have mutations in the *LDLR* gene that encodes the LDL receptor protein, which normally removes LDL from the circulation, or Apo B, which is the part of LDL that binds with the receptor.

In what food it is?

- Product of animal metabolism

meat **brain** **egg yolk** **liver**

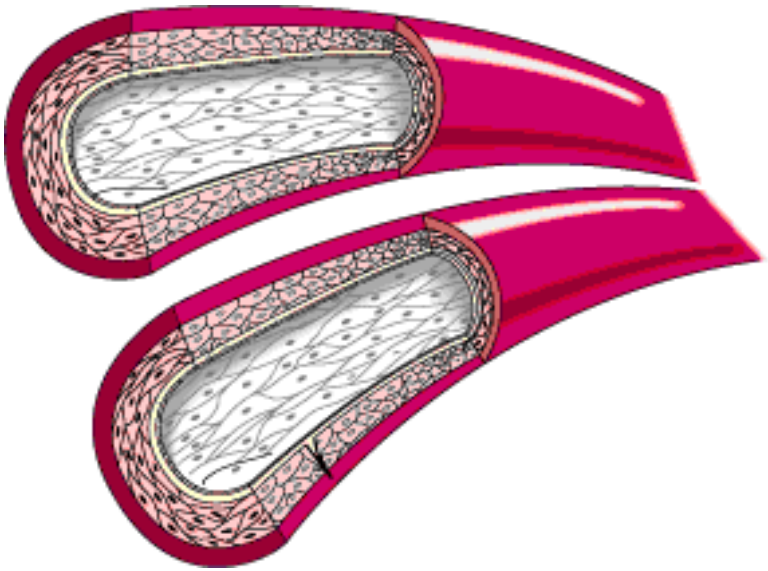
- It is risky to use technology which leads to heating up cholesterol with air (desiccation of milk and eggs, rendering of butter and lard – žluknutí – butter and lard turning rancid)

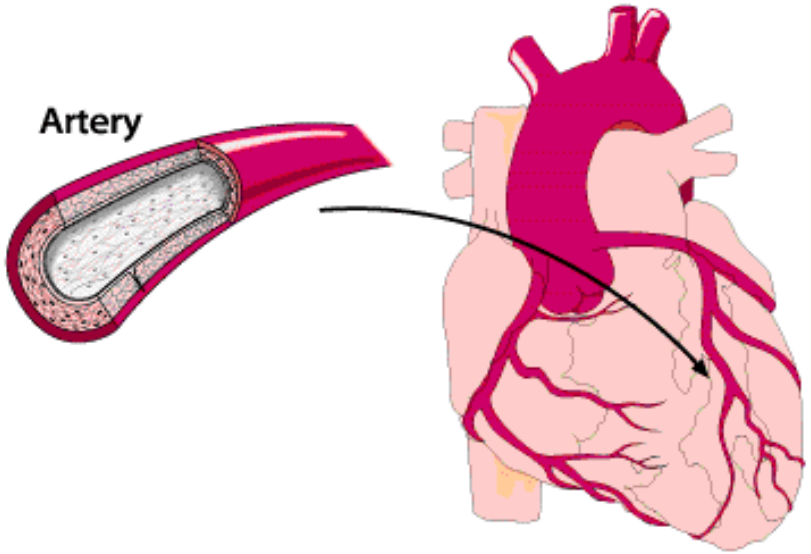
Where can it be found in body?

- free, or in form of cholesterol esters with fatty acids
- Their transport is allowed in blood only in form of specific elements - lipoproteins, in which fats are bound on proteins transmiters in combination with tryglicerides.

Diseases which are caused due to cholesterol

- cholesterol stores into vessels in the form of it's compounds – necking of vessels, ISCH, aterioscleros is





- Amount of cholesterol - optimal 300 mg per day and less.

Genetics and it´s role

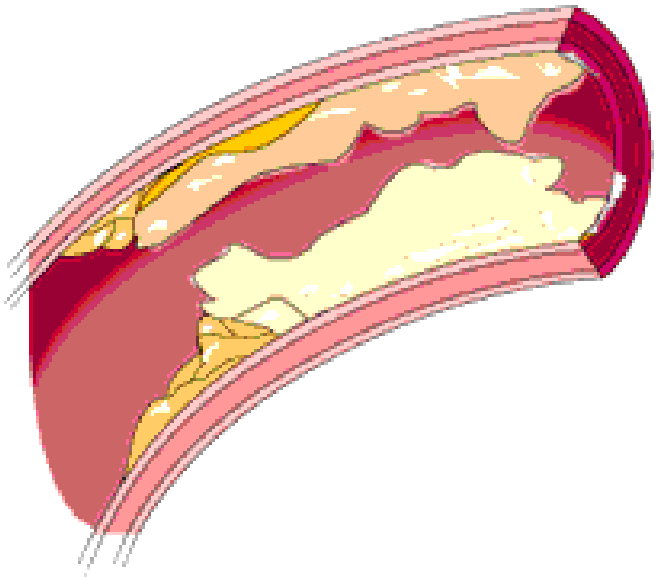
(look at our parents)

- Synthesis and degradation of cholesterol is genetically predicted.
- But the main influence have also outer conditions, mainly the food we eat.



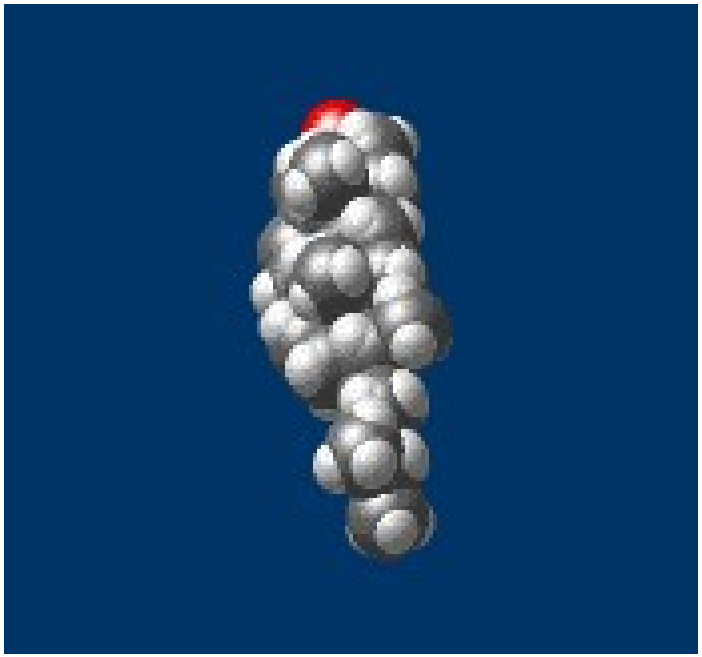
How to reduce cholesterol?

- Lower also income of sugars and maily alcohol
- medicaments: statines, fibrates and their combination



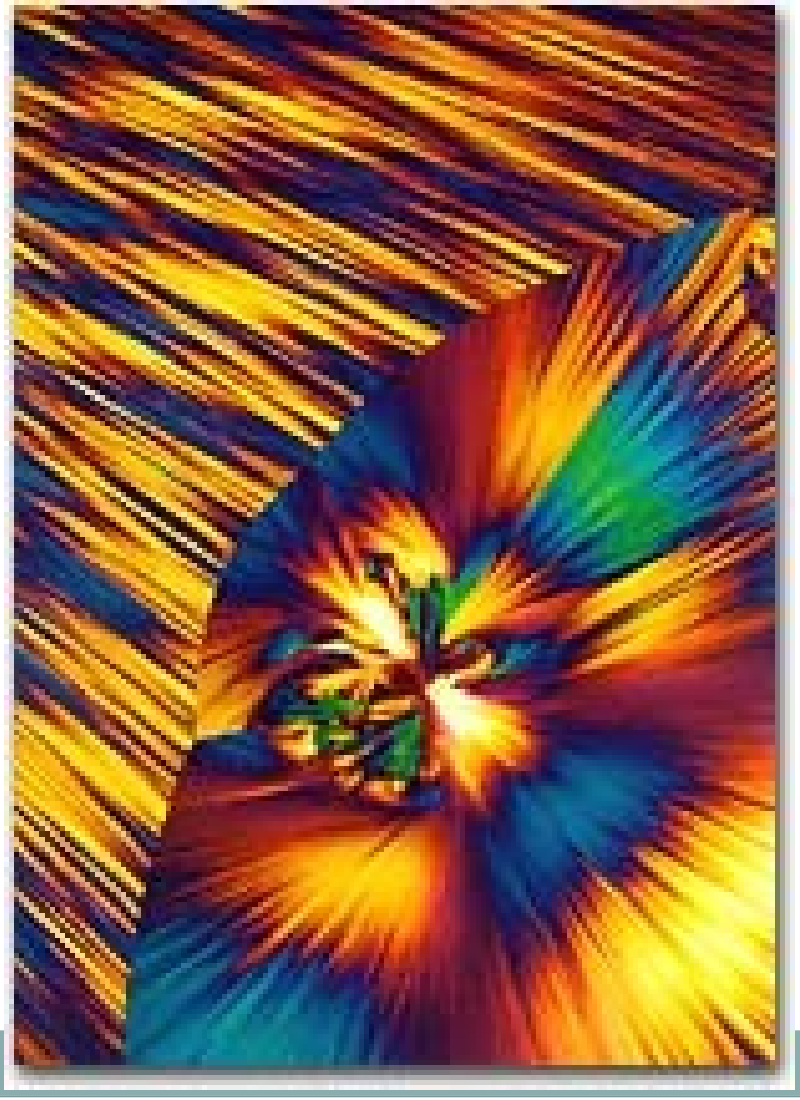
Curiosities

- For the first time cholesterol was isolated in 18th century from bile stones.
- Another research found that this steroid alcohol occurs in all cells of human body.



Curiosities

- Cholesterol under microscope



References

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